# MSG No. 188.219 $P\bar{6}c'2'$ [ Type III, hexagonal ]

Table 1: Wyckoff site: `2a`, site symmetry: `3.2'`

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, 0]$ | `[1,2,3,7,8,9]` |
| 2 | $\left[0, 0, \frac{1}{2}\right]$ | `[4,5,6,10,11,12]` |

Table 2: Wyckoff site: `2b`, site symmetry: `-6..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, 0, \frac{1}{4}\right]$ | `[1,2,3,4,5,6]` |
| 2 | $\left[0, 0, \frac{3}{4}\right]$ | `[7,8,9,10,11,12]` |

Table 3: Wyckoff site: `2c`, site symmetry: `3.2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{3}, \frac{2}{3}, 0\right]$ | `[1,2,3,7,8,9]` |
| 2 | $\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$ | `[4,5,6,10,11,12]` |

Table 4: Wyckoff site: `2d`, site symmetry: `-6..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$ | `[1,2,3,4,5,6]` |
| 2 | $\left[\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right]$ | `[7,8,9,10,11,12]` |

Table 5: Wyckoff site: `2e`, site symmetry: `3.2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{2}{3}, \frac{1}{3}, 0\right]$ | `[1,2,3,7,8,9]` |
| 2 | $\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$ | `[4,5,6,10,11,12]` |

Table 6: Wyckoff site: `2f`, site symmetry: `-6..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$ | `[1,2,3,4,5,6]` |
| 2 | $\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$ | `[7,8,9,10,11,12]` |

Table 7: Wyckoff site: `4g`, site symmetry: `3..`

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, z]$ | `[1,2,3]` |
| 2 | $\left[0, 0, \frac{1}{2} - z\right]$ | `[4,5,6]` |

*continued ...*

Table 7

| No. | position | mapping |
|---|---|---|
| 3 | $[0,\ 0,\ -z]$ | `[7,8,9]` |
| 4 | $\left[0,\ 0,\ z+\frac{1}{2}\right]$ | `[10,11,12]` |

Table 8: Wyckoff site: `4h`, site symmetry: `3..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$ | `[1,2,3]` |
| 2 | $\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$ | `[4,5,6]` |
| 3 | $\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$ | `[7,8,9]` |
| 4 | $\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{2}\right]$ | `[10,11,12]` |

Table 9: Wyckoff site: `4i`, site symmetry: `3..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$ | `[1,2,3]` |
| 2 | $\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}-z\right]$ | `[4,5,6]` |
| 3 | $\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$ | `[7,8,9]` |
| 4 | $\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$ | `[10,11,12]` |

Table 10: Wyckoff site: `6j`, site symmetry: `..2'`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ -x,\ 0]$ | `[1,9]` |
| 2 | $[x,\ 2x,\ 0]$ | `[2,7]` |
| 3 | $[-2x,\ -x,\ 0]$ | `[3,8]` |
| 4 | $\left[-2x,\ -x,\ \frac{1}{2}\right]$ | `[4,10]` |
| 5 | $\left[x,\ -x,\ \frac{1}{2}\right]$ | `[5,11]` |
| 6 | $\left[x,\ 2x,\ \frac{1}{2}\right]$ | `[6,12]` |

Table 11: Wyckoff site: `6k`, site symmetry: `m..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ y,\ \frac{1}{4}\right]$ | `[1,5]` |
| 2 | $\left[-y,\ x-y,\ \frac{1}{4}\right]$ | `[2,6]` |
| 3 | $\left[-x+y,\ -x,\ \frac{1}{4}\right]$ | `[3,4]` |
| 4 | $\left[x,\ x-y,\ \frac{3}{4}\right]$ | `[7,12]` |
| 5 | $\left[-x+y,\ y,\ \frac{3}{4}\right]$ | `[8,10]` |
| 6 | $\left[-y,\ -x,\ \frac{3}{4}\right]$ | `[9,11]` |

Table 12: Wyckoff site: `12l`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | `[1]` |
| 2 | $[-y,\ x-y,\ z]$ | `[2]` |
| 3 | $[-x+y,\ -x,\ z]$ | `[3]` |
| 4 | $\left[-x+y,\ -x,\ \frac{1}{2}-z\right]$ | `[4]` |
| 5 | $\left[x,\ y,\ \frac{1}{2}-z\right]$ | `[5]` |
| 6 | $\left[-y,\ x-y,\ \frac{1}{2}-z\right]$ | `[6]` |
| 7 | $[x,\ x-y,\ -z]$ | `[7]` |
| 8 | $[-x+y,\ y,\ -z]$ | `[8]` |
| 9 | $[-y,\ -x,\ -z]$ | `[9]` |
| 10 | $\left[-x+y,\ y,\ z+\frac{1}{2}\right]$ | `[10]` |
| 11 | $\left[-y,\ -x,\ z+\frac{1}{2}\right]$ | `[11]` |
| 12 | $\left[x,\ x-y,\ z+\frac{1}{2}\right]$ | `[12]` |